

SQL Code

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1          -- =====
2          -- CLIMATE CHANGE ANALYSIS PROJECT
3          -- =====
4
5      -- =====
6      -- DATABASE CREATION
7      -- =====
8      CREATE DATABASE climate_data;
9
10
11     -- =====
12     -- SELECT DATABASE
13     -- =====
14     USE climate_data;
15
16     -- =====
17     -- VIEWING TABLES
18     -- =====
19     show tables;
20
21     -- =====
22     -- 4. JOINING DATASETS USING YEAR AND REGION
23     -- =====
24
25     -- ==== Q-->2. Write queries to join datasets on Year + Region.
26     SELECT t.Year,t.Country,t.Region, t.`Temp_Anomaly °C`, c.`CO2_Emissions
27     (MtCO2)`,r.Rainfall_mm FROM temperature_anomalies t
28     JOIN co2_emissions c ON t.Year = C.Year AND t.Country = C.country
29     JOIN rainfall r ON t.Year=r.year AND t.Country=r.country
30     ORDER BY t.Year=t.Country;
31
32     -- =====
33     -- AVERAGE TEMPERATURE ANOMALIES BY CONTINENT
34     -- =====
35
36     -- ==== Q->3. Use GROUP BY to calculate average anomalies per continent.
37
38     SELECT t.Region,ROUND(AVG(t.`Temp_Anomaly °C`),2) AS Avg_Temp_Anomaly,
39     ROUND(AVG(c.`CO2_Emissions (MtCO2)`),2) AS Avg_CO2_Emissions,
40     ROUND(AVG(r.Rainfall_mm),2) AS AVG_Rainfall FROM temperature_anomalies t
41     JOIN co2_emissions c ON t.Year=C.Year AND t.Country=C.Country
42     JOIN Rainfall r ON t.Year=r.Year AND t.Country =r.Country
43     GROUP BY t.Region ORDER BY Avg_Temp_Anomaly DESC;
44
45     -- =====
46     -- CUMULATIVE CO2 EMISSIONS USING WINDOW FUNCTION
47     -- =====
48     -- ====Q->4. Apply Window Functions for cumulative CO2 emissions.

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49 SELECT Year, Country, Region, 'CO2_Emissions (MtCO2)', SUM(`CO2_Emissions (MtCO2)`) OVER
(PARTITION BY Country ORDER BY Year
50 ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW) AS Cumulative_Co2
51 FROM Co2_emissions ORDER BY Country, Year;
52
53 -- =====
54 -- TOP 5 REGIONS WITH HIGHEST ANOMALIES
55 -- =====
56
57 -- ==== Q->5. Write a query to find top 5 regions with highest anomalies.
58 SELECT Country, Region, ROUND(AVG(`Temp_Anomaly °C`), 2) AS Avg_Temp_Anomaly FROM
temperature_anomalies
59 GROUP BY Country, Region ORDER BY Avg_Temp_Anomaly DESC LIMIT 5;
60
61 -- ==== Q->6. Use CASE WHEN to categorize emission levels.
62 SELECT
63     Year,
64     Country,
65     Region,
66     `CO2_Emissions (MtCO2)`,
67     CASE
68         WHEN `CO2_Emissions (MtCO2)` < 200 THEN 'Low'
69         WHEN `CO2_Emissions (MtCO2)` < 600 THEN 'Medium'
70         WHEN `CO2_Emissions (MtCO2)` < 1500 THEN 'High'
71         ELSE 'Critical'
72     END AS Emission_Category
73 FROM co2_emissions
74 ORDER BY `CO2_Emissions (MtCO2)` DESC;
75
76 -- =====
77 -- STORED PROCEDURE FOR YEARLY CLIMATE SUMMARY
78 -- =====
79
80 -- ==== Q->7. Create a stored procedure for yearly climate summary.
81
82 USE climate_data;
83
84 DELIMITER $$
85 CREATE PROCEDURE yearly_climate_summary(IN input_year INT)
86 BEGIN
87     SELECT
88         t.Year, t.Region,
89         ROUND(AVG(t.`Temp_Anomaly °C`), 2) AS Avg_Temp_Anomaly,
90         ROUND(SUM(c.`CO2_Emissions (MtCO2)`, 2) AS Total_Co2,
91         ROUND(AVG(r.Rainfall_mm), 2) AS AVG_Rainfall
92     FROM temperature_anomalies t
93     JOIN co2_emissions c ON t.Year=c.Year AND t.Country=c.Country
94     JOIN rainfall r ON t.Year =r.Year AND t.Country=r.Country
95     WHERE t.Year = input_year
96     GROUP BY t.Year, t.Region
97     ORDER BY Avg_Temp_Anomaly DESC;
98 END$$
99 DELIMITER ;
100 CALL yearly_climate_summary (2015);

```

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101
102 -- =====
103 -- VIEW CREATION FOR RENEWABLE ENERGY TRENDS
104 -- =====
105 -- ==== Q->8. Build a view for renewable energy adoption trends.
106 CREATE VIEW renewable_energy_trends AS
107 SELECT Year,Region,`Renewable_Energy (%)` FROM renewable_energy;
108 SELECT * FROM renewable_energy_trends;
109 -- ===== From this we can check my view
110
111 -- =====
112 -- DETECTING MISSING RAINFALL RECORDS
113 -- =====
114
115 -- ==== Q->9. Write a query to detect missing rainfall records.
116 USE climate_data;
117 SELECT Year,Country,Region,Rainfall_mm FROM Rainfall WHERE Rainfall_mm IS NULL OR
Rainfall_mm=0 ORDER BY year,region;
118
119 -- =====
120 -- EXPORTING FINAL DATASET
121 -- =====
122 -- ==== Q->10 Export SQL results into CSV for Python analysis.
123 -- EXPORTING Temperature DATASET
124 -- =====
125 USE climate_data;
126 SELECT
127     'year',
128     'region',
129     'country',
130     'Temp_Anomaly °C)'
131 UNION ALL
132 SELECT
133     year,
134     region,
135     country,
136     `Temp_Anomaly °C)`
137 FROM temperature_anomalies
138 INTO OUTFILE 'C:/ProgramData/MySQL/MySQL Server 9.2/Uploads/temperature.csv'
139 FIELDS TERMINATED BY ','
140 ENCLOSED BY '"'
141 LINES TERMINATED BY '\n';
142
143 -- EXPORTING CO2 Emissions CSV
144 -- =====
145
146 SELECT
147     'year',
148     'region',
149     'country',
150     'CO2_Emissions (MtCO2)'
151 UNION ALL
152 SELECT
153     year,

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154     region,
155     country,
156     `CO2_Emissions (MtCO2)`
157 FROM CO2_emissions
158 INTO OUTFILE 'C:/ProgramData/MySQL/MySQL Server 9.2/Uploads/co2_emissions.csv'
159 FIELDS TERMINATED BY ','
160 ENCLOSED BY '"'
161 LINES TERMINATED BY '\n';
162
163 -- EXPORTING Rainfall CSV
164 -- =====
165 SELECT
166     'year',
167     'region',
168     'country',
169     'rainfall_mm'
170 UNION ALL
171 SELECT
172     year,
173     region,
174     country,
175     rainfall_mm
176 FROM Rainfall
177 INTO OUTFILE 'C:/ProgramData/MySQL/MySQL Server 9.2/Uploads/rainfall.csv'
178 FIELDS TERMINATED BY ','
179 ENCLOSED BY '"'
180 LINES TERMINATED BY '\n';
181
182
183 SHOW VARIABLES LIKE 'secure_file_priv';
```